SREBF2 (CHR22)

Gene Report

Prepared for the CHDI Foundation
by Rancho BioSciences

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Table of Contents

1. General Gene Information

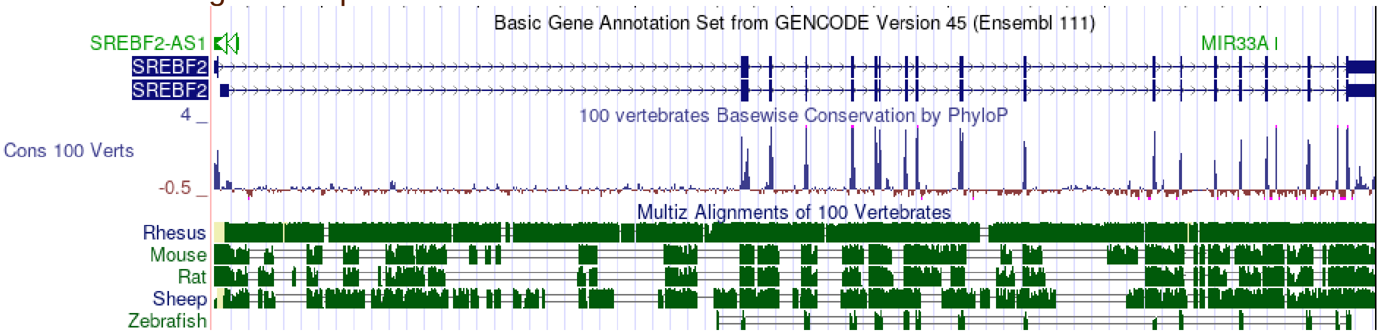
a. Gene Aliases

	Human gene	Mouse gene	Rat gene
Entrez Gene ID	6721	20788	499505
Gene Symbol	SREBF2	Srebf2	Srebf2
Gene Name	sterol regulatory element binding transcription factor 2	sterol regulatory element binding factor 2	sterol regulatory element binding transcription factor 2
Ensembl ID	ENSG00000198911	ENSMUSG00000022463	ENSRNOG00000007400
Uniprot ID	Q12772	Q3U1N2	Q3T1I5
Synonyms/ Aliases	BHLHD2, SREBP2	SREBP-2, SREBP2, SREBP2gc, bHLHd2, lop13, nuc	SREBP2; SREBP-2; Srebf2_retired

b. Conservation of gene across species and conservation of region under peak in other genomes

The UCSC Genome browser hosts the complete human genome alongside different versions of human genome assembly and includes a large collection of vertebrate and model organism assemblies and annotations. This tool provides a graphical interface for viewing, analyzing, and downloading data. A query of human Dec. 2013 (GRCh38/hg38) in Genome browser found 51 results for SREBF2 under the "GENCODE V44" tracks. Conservation Track of one transcript was shown below:

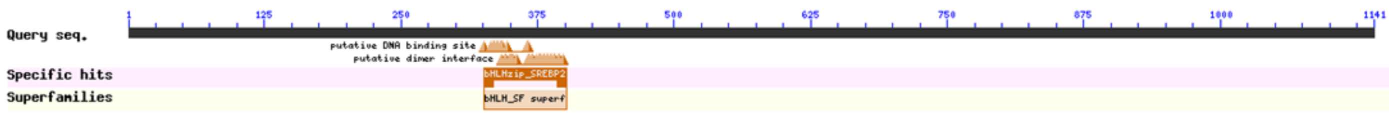
SREBF2 (ENST00000361204.9) - chr22:41833105-41907305 - Homo sapiens sterol regulatory element binding transcription factor



c. Known structure

The NCBI Structure Group and Conserved Domain Database (CDD) are protein annotation resources utilizing multiple sequence alignment models for identification of conserved domains in protein sequences. The RCSB Protein Data Bank (PDB) is another open resource for assessing protein structure, providing access to 3D structure data for large biological molecules (proteins, DNA, and RNA). Below is a subset of the most relevant domains and structures, with links to additional information.

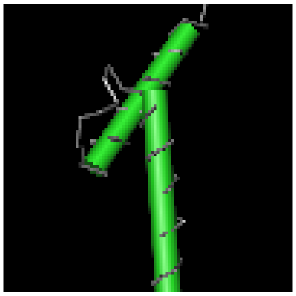
Conserved domains (NCBI CDD link):



basic Helix-Loop-Helix-zipper (bHLHzip) domain found in sterol regulatory element-binding protein 2 (SREBP2) and similar proteins:

Conserved Protein Domain Family bHLH_SF:

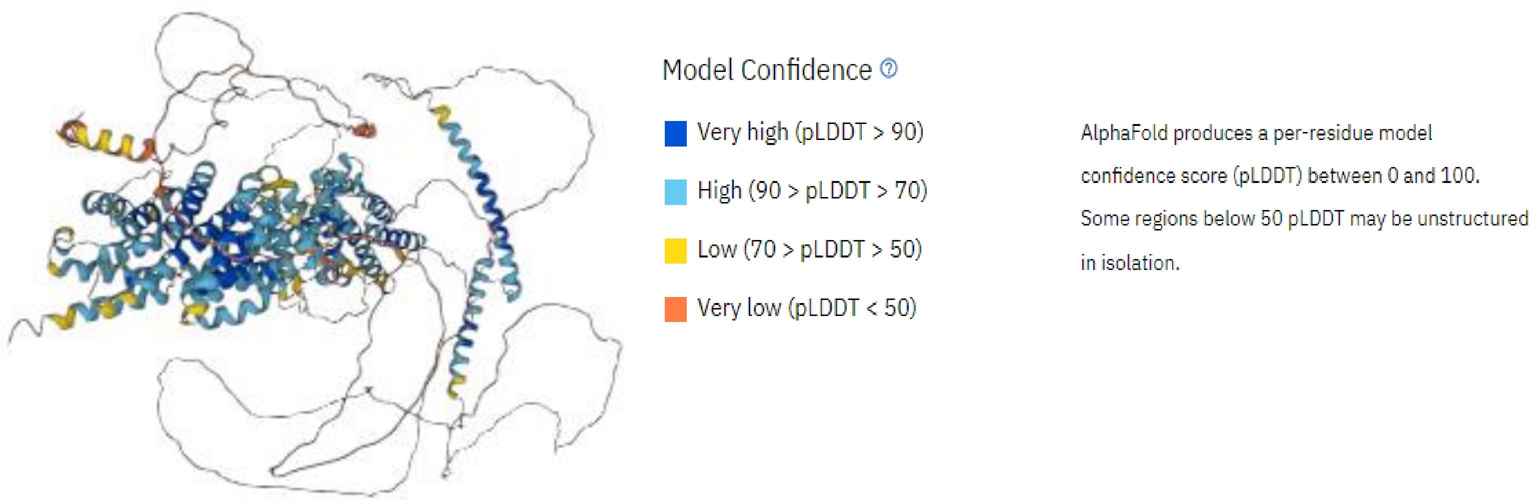
basic Helix Loop Helix (bHLH) domain superfamily: bHLH proteins are transcriptional regulators that are found in organisms from yeast to humans. Members of the bHLH superfamily have two highly conserved and functionally distinct regions. The basic part is at the amino end of the bHLH that may bind DNA to a consensus hexanucleotide sequence known as the E box (CANNTG). Different families of bHLH proteins recognize different E-box consensus sequences. At the carboxyl-terminal end of the region is the HLH region that interacts with other proteins to form homo- and heterodimers. bHLH proteins function as a diverse set of regulatory factors because they recognize different DNA sequences and dimerize with different proteins. The bHLH proteins can be divided to cell-type specific and widely expressed proteins. The cell-type specific members of bHLH superfamily are involved in cell-fate determination and act in neurogenesis, cardiogenesis, myogenesis, and hematopoiesis.



3D structures from PDB: Human SREBF2 protein (PDB link)

d. AlphaFold protein structure prediction

Human SREBF2: AlphaFold Protein Structure Link



Rat Srebf2: AlphaFold Protein Structure Link

Mouse Srebf2: AlphaFold Protein Structure Link

e. Homologues in model animals

NCBI HomoloGene provides integrated information regarding gene nomenclature, reference sequences, transcripts, orthologs, pathways, and phenotypes from a wide range of species. There are 562 genes that are orthologs to human SREBF2 in jawed vertebrates. The SREBF2 gene is conserved in mouse, rat, zebrafish, chicken, Chinese hamster, pig, cattle, frog, monkey, chimpanzee, rabbit, etc. NCBI Orthologs for human SREBF2 (Link) are shown below and additional orthologs available in OrthoDB (Link).

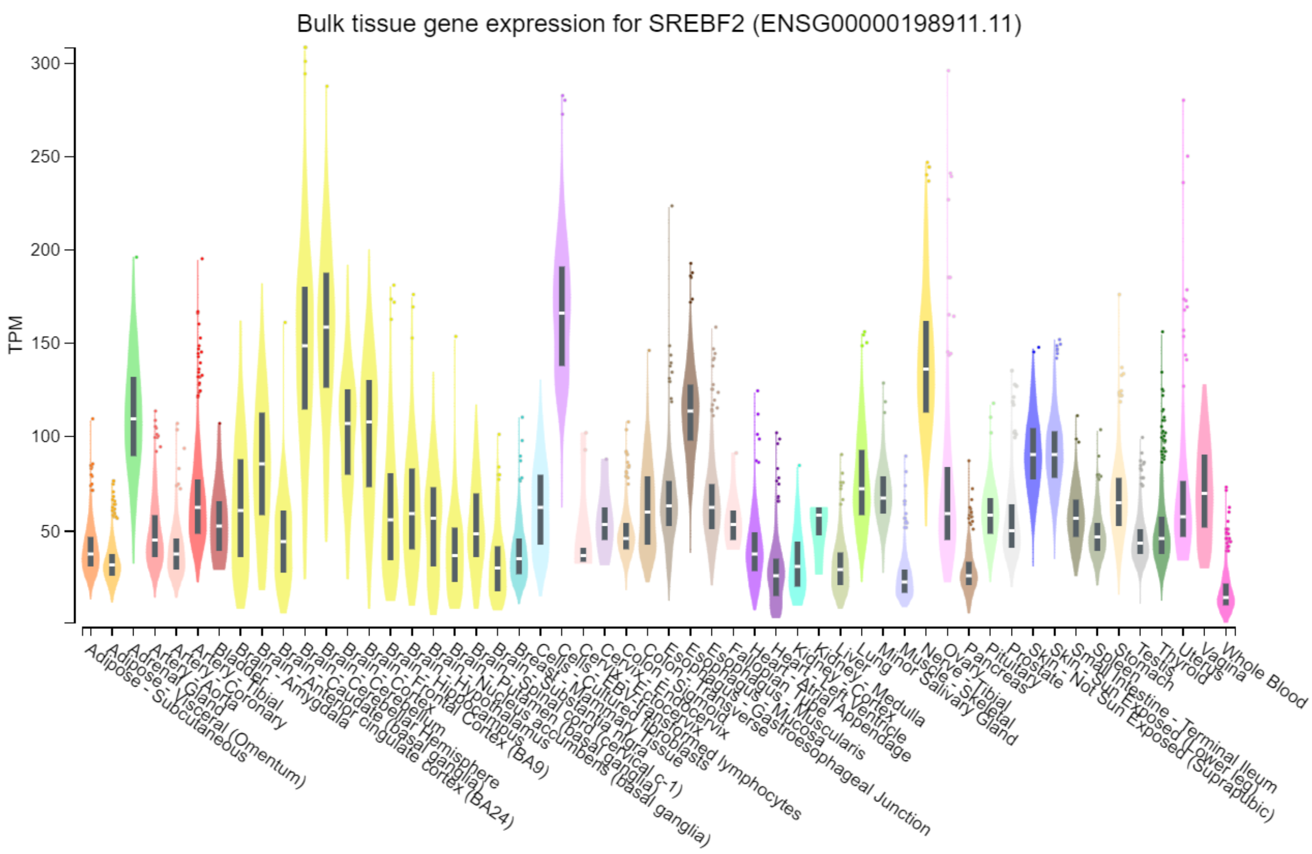
Species	Gene	Architecture	aa
Homo sapiens human	SREBF2 sterol regulatory element binding transcription factor 2		1,141
Mus musculus house mouse	Srebf2 sterol regulatory element binding factor 2		1,130
Rattus norvegicus Norway rat	Srebf2 sterol regulatory element binding transcription factor 2		1,133
Danio rerio zebrafish	srebf2 sterol regulatory element binding transcription factor 2		1,099
Gallus gallus chicken	SREBF2 sterol regulatory element binding transcription factor 2		1,112
Cricetulus griseus Chinese hamster	Srebf2 sterol regulatory element binding transcription factor 2		1,139
Sus scrofa pig	SREBF2 sterol regulatory element binding transcription factor 2		1,135
Bos taurus cattle	SREBF2 sterol regulatory element binding transcription factor 2		1,140
Xenopus tropicalis tropical clawed frog	srebf2 sterol regulatory element binding transcription factor 2		1,096
Macaca mulatta Rhesus monkey	SREBF2 sterol regulatory element binding transcription factor 2		1,248

2. Expression pattern by cell and tissue

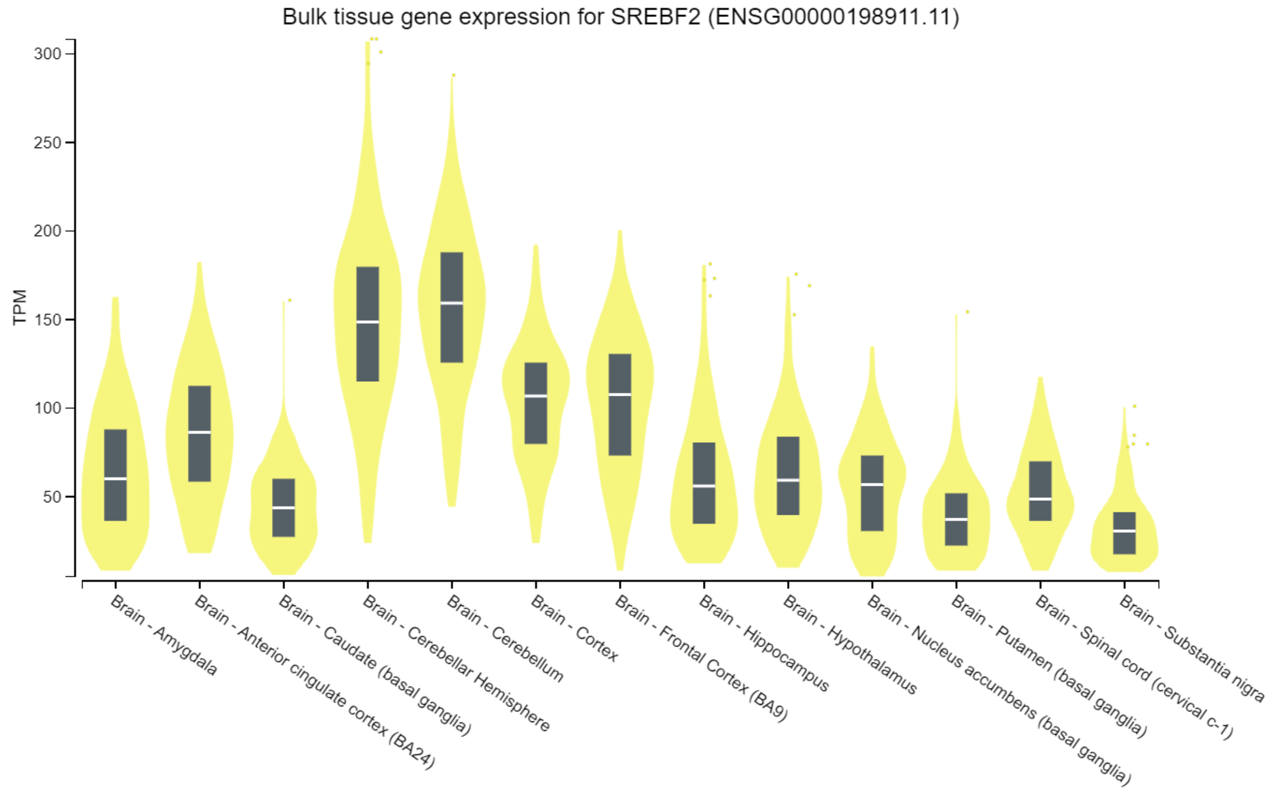
a. Tissue-specific information

The Genotype-Tissue Expression (GTEx) project is a comprehensive public resource that provides tissue-specific gene expression and regulation data, including RNA-Seq, eQTLs, and histology images. The resource contains information collected from 53 non-diseased tissue sites across nearly 1000 individuals, utilizing primarily molecular assays such as whole genome/exome sequencing and RNA-Seq. Below is information pertaining to SREBF2 expression across all tissues and specifically in the brain.

GTEx: SREBF2 expression in all tissues

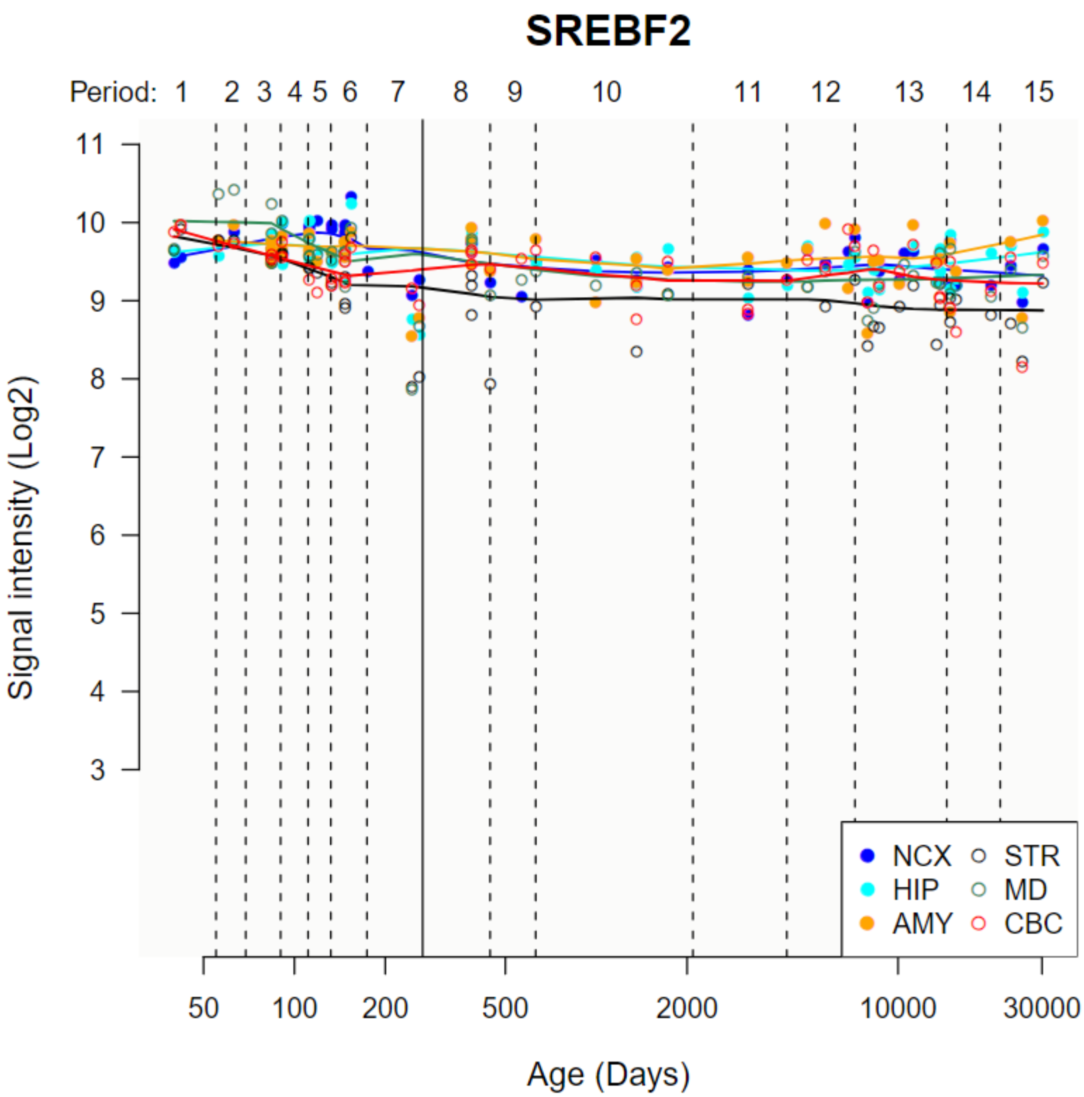


GTEx: SREBF2 expression in brain



The Human Brain Transcriptome (HBT) is a public database hosted by the Department of Neurobiology at Yale University School of Medicine. It contains transcriptome data and associated metadata from over 1,340 tissue samples, specifically focusing on the developing and adult human brain.

SREBF2 gene expression (Link) along entire development and adulthood in the cerebellar cortex (CBC), mediodorsal nucleus of the thalamus (MD), striatum (STR), amygdala (AMY), hippocampus (HIP) and 11 areas of neocortex (NCX) is shown below.



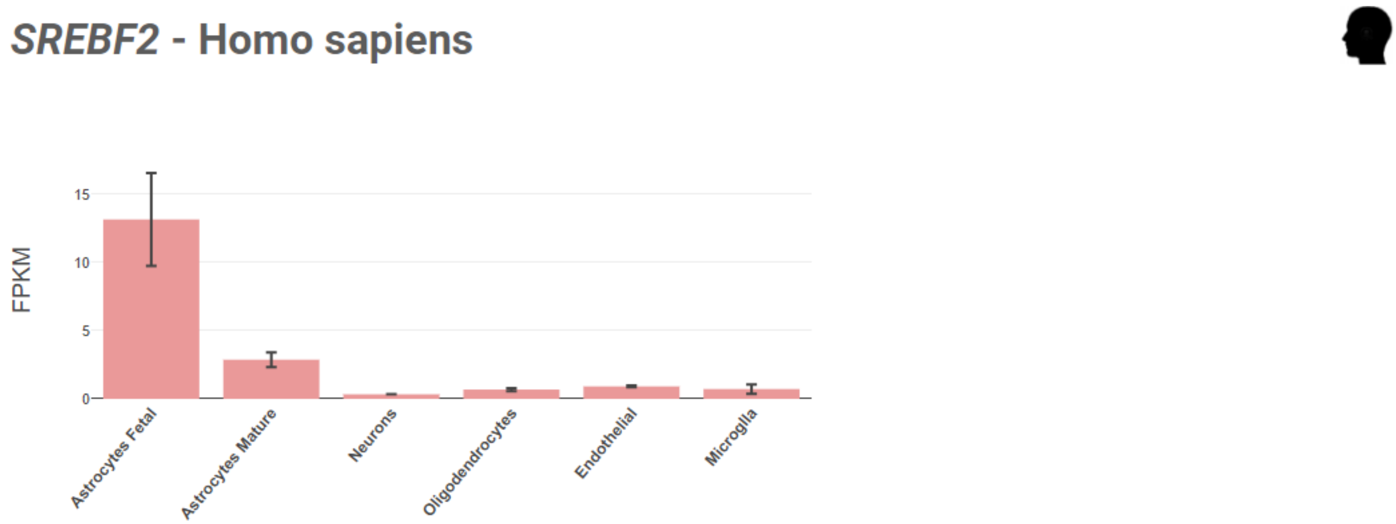
b. Barres Lab RNA-Seq brain specific expression data

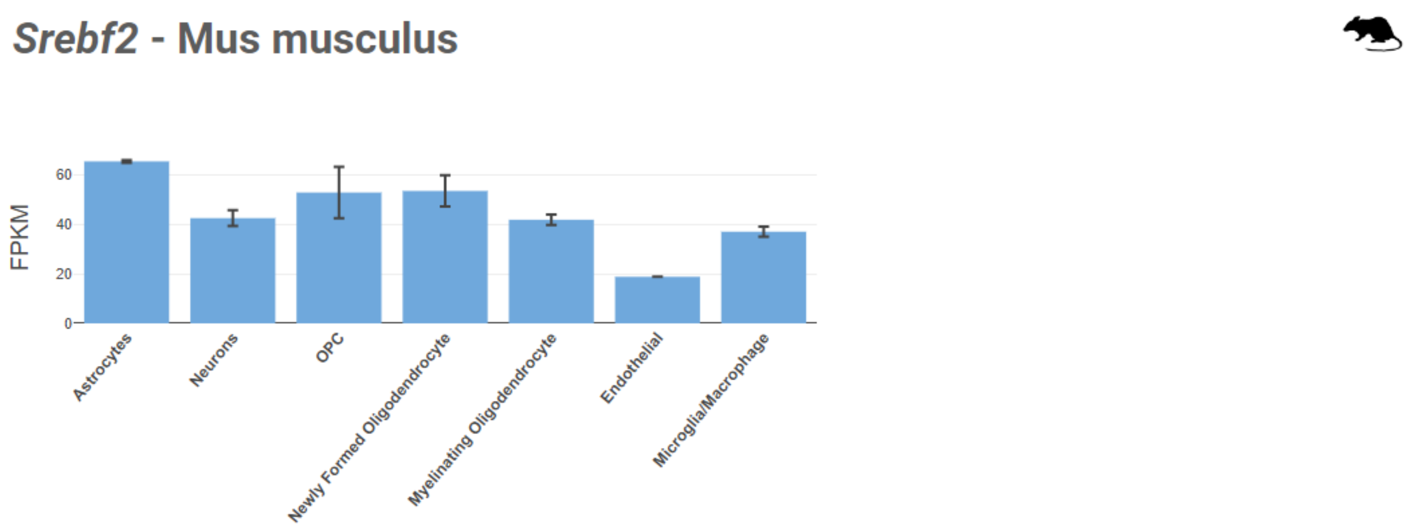
Expression data from Allen Brain Atlas and Barres Lab RNA-Seq were downloaded and average expression levels were calculated. The SREBF2 expression results are presented in the table below.

Average Human Brain Agilent Expression	Average Human Brain RNA-Seq Expression (TPM)	Average Mouse Brain RNA-Seq Expression (TPM)	Human Brain Expression Category	Mouse Brain Expression Category
7.023964492	176.6800531	199.8909211	medium	high

Additionally, relative SREBF2 RNA expression in various cell types isolated from human and mouse brain as measured by RNA-Seq is shown here, where FPKM is Fragments Per Kilobase Million, a normalization unit commonly utilized in this types of assay (Barres Lab Brain RNA-Seq).

Brain RNA-Seq generated by the Ben Barres lab (Link)



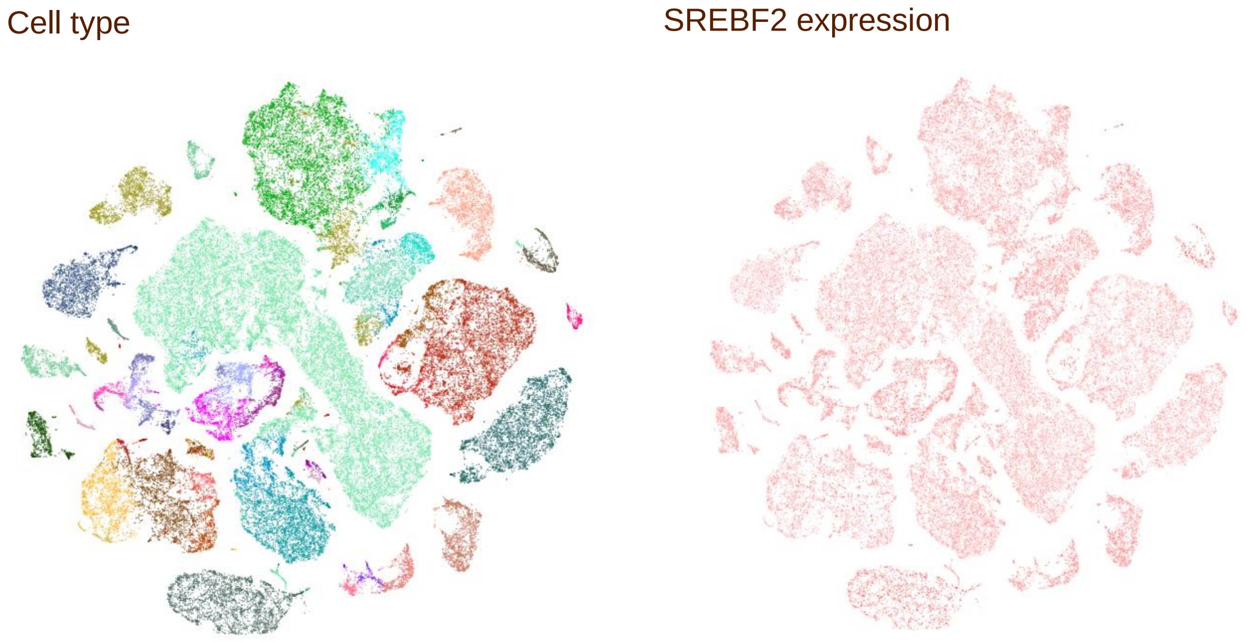


c. snRNA-Seq gene expression in cell type database

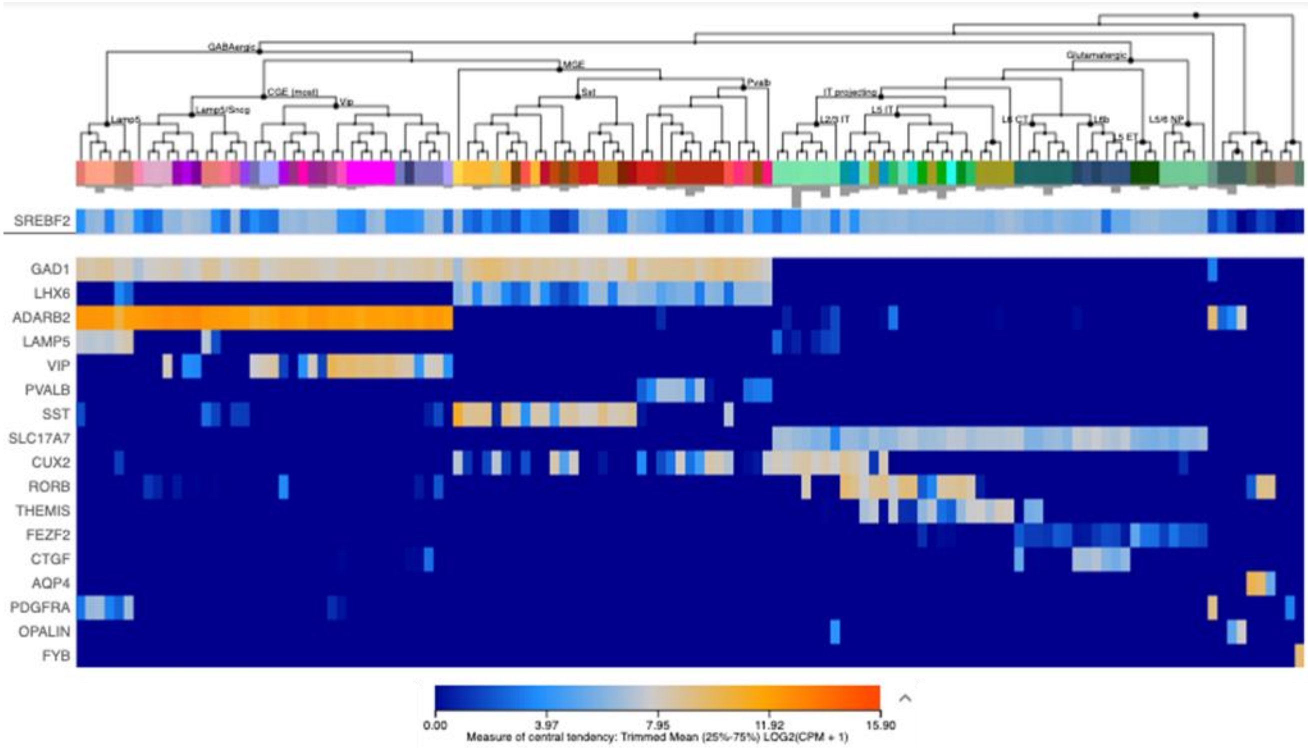
1. Single cell (or single nucleus) RNA sequencing (RNA-Seq) is a scalable approach to provide genome-wide expression profiles for thousands of cells. This data set includes single cell and nuclear transcriptomic profiles, assayed from human and mouse brain regions. Anatomical specificity is achieved by microdissecting tissue from defined brain areas, such as cortical layers or cell groups in LGd. Individual layers of cortex were dissected from tissues covering the middle temporal gyrus (MTG), anterior cingulate gyrus (CgGr), primary visual cortex (V1C), primary motor cortex (M1C),

primary somatosensory cortex (S1C) and primary auditory cortex (A1C) derived from human brain, and nuclei were dissociated and sorted using the neuronal marker NeuN. Nuclei were sampled from postmortem and neurosurgical (MTG only) donor brains and expression was profiled with SMART-Seq v4 or 10x Genomics Chromium Single Cell 3' v3 RNA-sequencing. Analysis for gene of interest in the Cell Types Database for selected datasets were shown below:

Human-M1-10x motor cortex dataset from Cell Type Database (Single cell RNA-seq data): For the Human-M1-10x motor cortex dataset, the left UMAP plot was overlaid and colored by cell sub-type. The large light green cluster in the middle represents the excitatory neurons from L2. The UMAP in the top right shows the expression of SREBF2 overlaid onto the UMAP. Each point represents a cell, and the darker the red points appear, the greater the expression of SREBF2. Based on the distribution, SREBF2 is broadly expressed at low expression levels across all cell types.

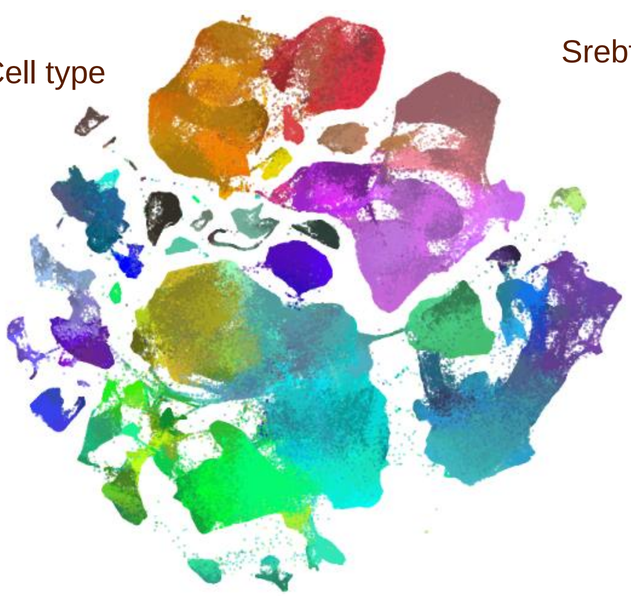


Human-M1-10x motor cortex dataset: gene expression heatmap (Single cell RNA-seq data): The heatmap below shows the expression of SREBF2 (first row) across all cell types in the Human-M1-10x motor cortex dataset as compared to several other genes. While GAD1 and SLC17A7 primarily underly cell type specific dendrogram construction, SREBF2 is broadly expressed at low expression levels across all cell types.

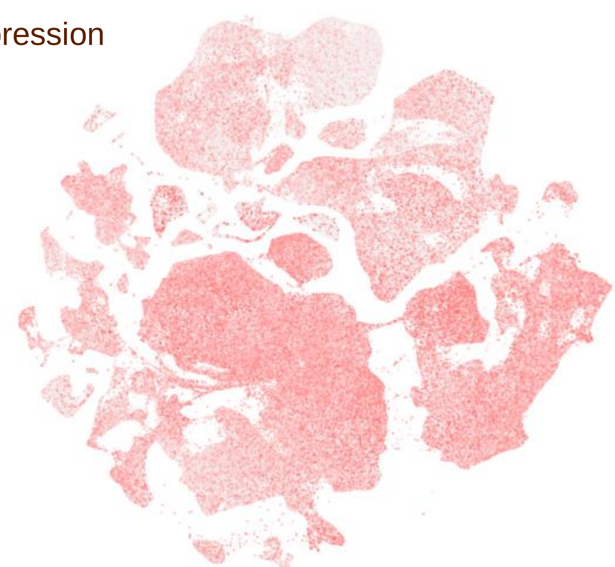


Mouse - WHOLE CORTEX & HIPPOCAMPUS - 10X GENOMICS (2020) dataset: For the Mouse – Whole Cortex and Hippocampus 10x dataset, the left UMAP plot was overlaid and colored by cell sub-type. The large bright red cluster at the top represents IN-Pvalb interneurons. The UMAP in the top right shows the expression of Srebf2 overlaid onto the UMAP. Each point represents a cell, and the darker the red points appear, the greater the expression of Srebf2. As is clear from the UMAP, the

Cell type

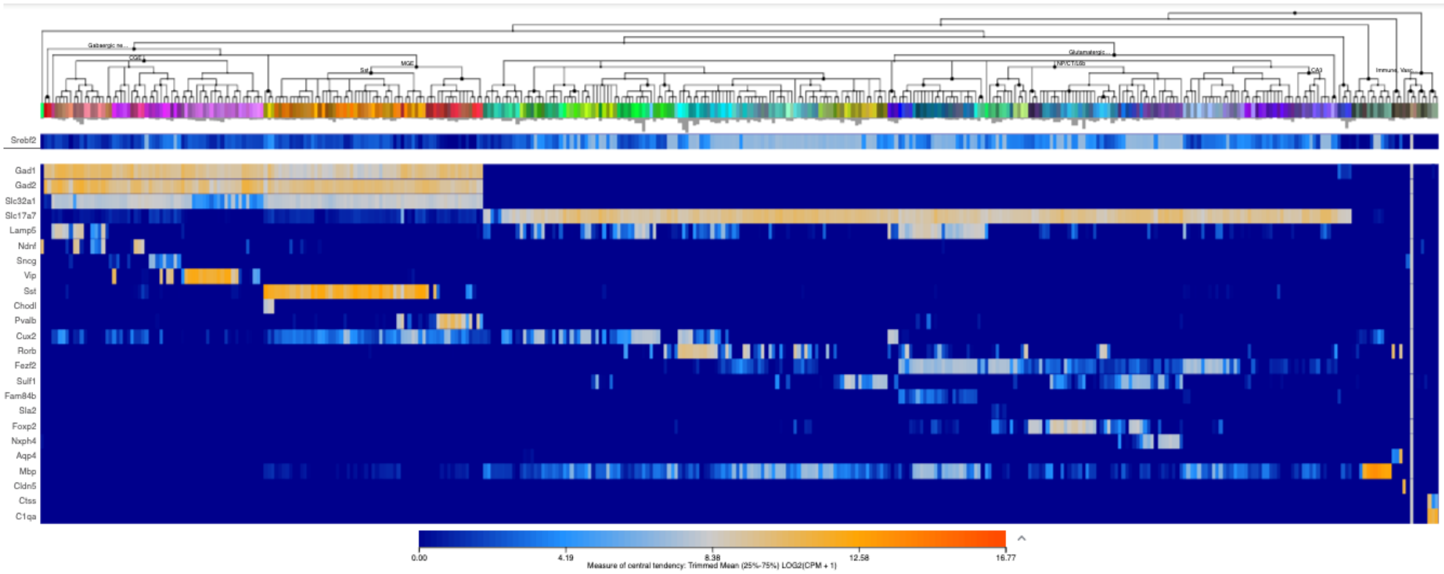


Srebf expression



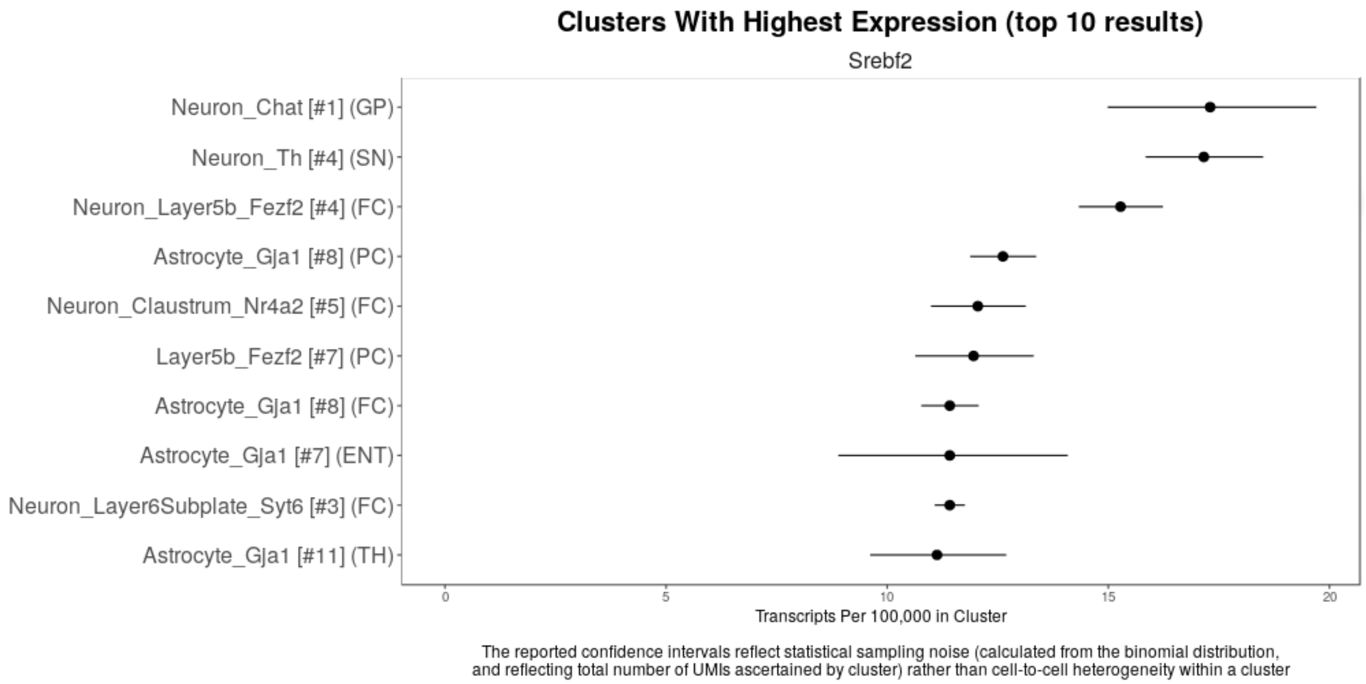
expression of Srebf2 is similar across all cell types, but with slightly higher expression in L6 IT, L5 IT and L4 IT.

Mouse - WHOLE CORTEX & HIPPOCAMPUS - 10X GENOMICS (2020) dataset: gene expression heatmap (Single cell RNA-seq data): For the Mouse – Whole Cortex and Hippocampus 10x dataset, the expression of Srebf2 (first row) is shown below across all cell types as compared to other genes. While Gad1, Gad2 and Slc17a7 primarily underly cell type specific dendrogram construction, Srebf2 expression is generally low across most cell types; however, similarly to the UMAP, there is greater expression of Srebf2 in L6 IT, L5 IT and L4 IT.

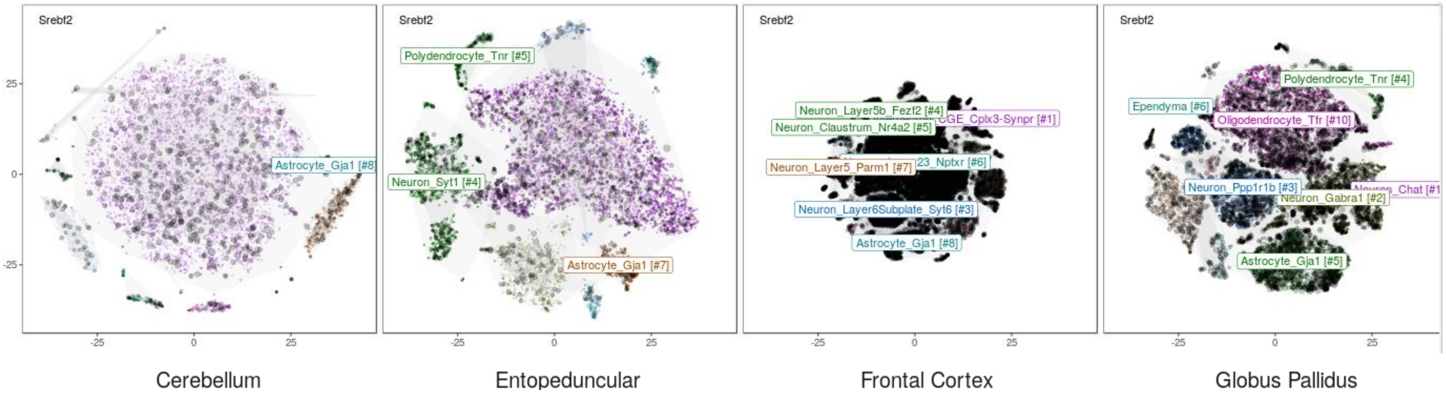


2. The mammalian brain is composed of diverse, specialized cell populations. To systematically ascertain and learn from these cellular specializations, we used Drop-seq to profile RNA expression in 690,000 individual cells sampled from 9 regions of the adult mouse brain. The value of DropViz is that it contains data from the entire mouse brain. We identified 565 transcriptionally distinct groups of cells using computational approaches developed to distinguish biological from technical signals. Cross-region analysis of these 565 cell populations revealed features of brain organization, including a gene-expression module for synthesizing axonal and presynaptic components, patterns in the co-deployment of voltage-gated ion channels, functional distinctions among the cells of the vasculature and specialization of glutamatergic neurons across cortical regions. Systematic neuronal classifications for two complex basal ganglia nuclei and the striatum revealed a rare population of spiny projection neurons. This adult mouse brain cell atlas, accessible through interactive online software (DropViz), serves as a reference for development, disease, and evolution.

Levels by cluster for Srebf2: The greatest expression of Srebf2 is found in the neurons cluster 1 (specifically, IN-Chat interneurons), Th cluster 4 and neuron layer 5b in cluster 4.



[tSNE analysis of global clusters for Srebf2]: The tSNE plot shows expression of Srefb2 expression overlaid onto UMAPs by brain region. The greatest expression of Srefb2 is found in the granular neurons in the cerebellum. In contrast, there is higher expression of Srefb2 in the oligodendrocytes of the brain regions of the Entopeduncular and Globus Pallidus. There is very little expression of Srefb2 in the frontal cortex.



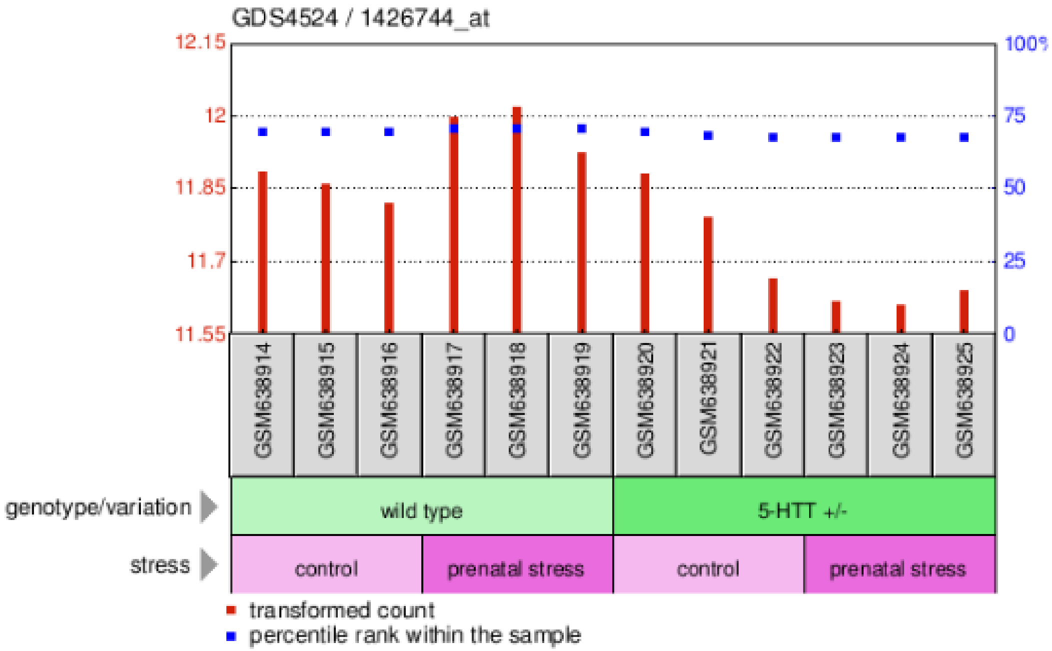
3. Perturbations in GEO that alter the gene

a. GEO Profiles search focusing on brain and/or neurons

The Gene Expression Omnibus (GEO) repository is a public database that stores MIAME (Minimum Information About a Microarray Experiment)- and MINSEQE (Minimum Information About a Next-generation Sequencing Experiment)-compliant gene expression data. GEO enables the search for pre-computed profiles, or a specific profile based on gene annotation. A search of the GEO Profiles database for datasets containing SREBF2 yielded 6505 datasets. Filtering for up/down genes decreased the number of datasets to 28.

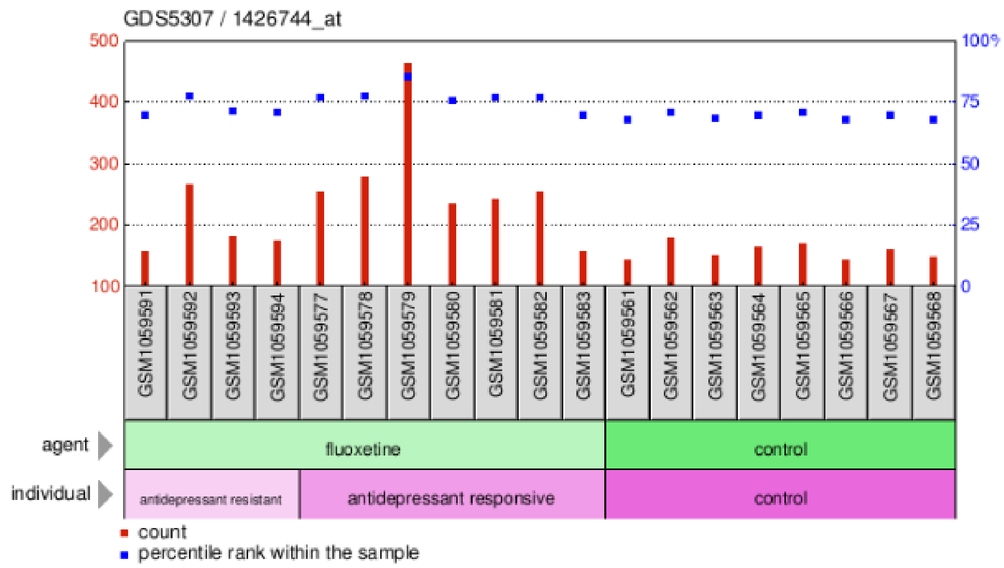
The following GEO profiles were selected as potentially relevant:

1. Srebf2 - Prenatal stress effect on serotonin transporter-deficient, female offspring: left hippocampus
Annotation: Srebf2, sterol regulatory element binding factor 2
Organism: Mus musculus
Reporter: GPL1261, 1426744_at (ID_REF), GDS4524, 20788 (Gene ID), BM123132
DataSet type: Expression profiling by array, transformed count, 12 samples
ID: 97740750

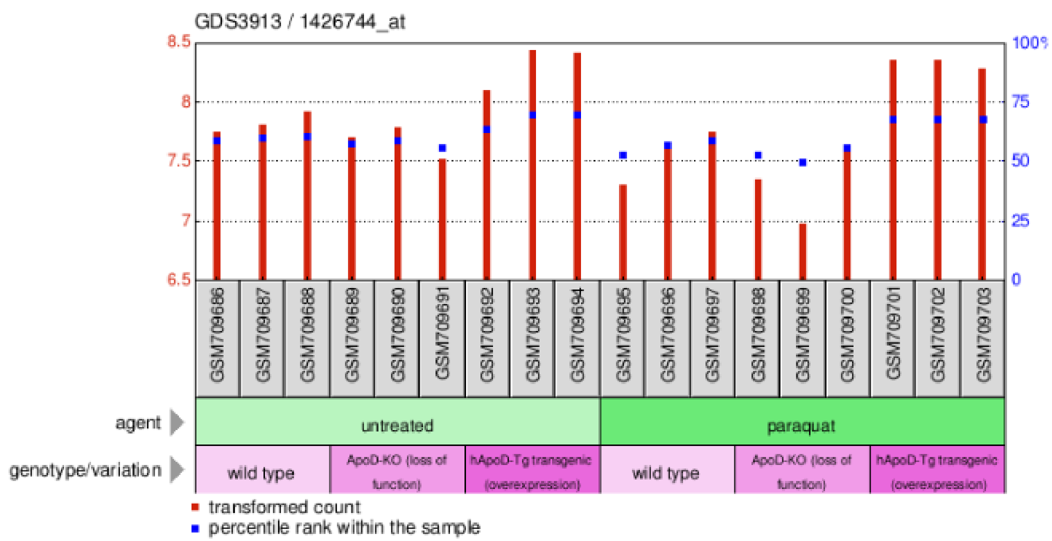


2. Srebf2 - Antidepressant fluoxetine effect on depression model: dorsal dentate gyrus
Annotation: Srebf2, sterol regulatory element binding factor 2
Organism: Mus musculus

Reporter: GPL1261, 1426744_at (ID_REF), GDS5307, 20788 (Gene ID), BM123132
DataSet type: Expression profiling by array, count, 19 samples
ID: 121571950



3. Srebf2 - Paraquat effect on Apolipoprotein D mutants: cerebellum
Annotation: Srebf2, sterol regulatory element binding factor 2
Organism: Mus musculus
Reporter: GPL8321, 1426744_at (ID_REF), GDS3913, 20788 (Gene ID), BM123132
DataSet type: Expression profiling by array, transformed count, 18 samples
ID: 72531839



4. Transcription factors that drive the gene's expression

a. Harmonizome Integrated Knowledge About Genes & Proteins

Transcription factors (TFs) are proteins responsible for modulating gene transcription. The Harmonizome is a collection of processed datasets gathered to serve and mine knowledge about genes and proteins. Multiple datasets have been integrated, including transcription factor targets and transcription factor binding site profiles from the following resources:

ENCODE (Encyclopedia of DNA Elements), a compilation of functional elements in the human genome, including transcription factors and other circumstances under which a gene is active.

ChEA (ChIP Enrichment Analysis), a gene-set enrichment analysis tool tailored to test if query gene-sets are enriched with genes that are putative targets of transcription factors.

MotifMap, which uses databases of transcription factor binding motifs, refined genome alignments, and a comparative genomic statistical approach to provide comprehensive maps of candidate regulatory elements encoded in the genomes of model species

JASPAR, an open-access database of curated, non-redundant transcription factor-binding profiles stored as position frequency matrices (PFMs) for TFs across multiple species in six taxonomic groups.

SREBF2 has 1028 transcription factor associations.

Association	Dataset	Tissue/Cells	Organism	Genome Build
TEAD4	ENCODE Transcription Factor Binding Site Profiles	HepG2		hg19_1
USF2	ENCODE Transcription Factor Binding Site Profiles	GM12878		hg19_1
TBP	ENCODE Transcription Factor Binding Site Profiles	HeLa-S3		hg19_1
SUZ12	ENCODE Transcription Factor Targets			
MAX	ENCODE Transcription Factor Targets			

Association	Dataset	Tissue/Cells	Organism	Genome Build
NFIC	ENCODE Transcription Factor Binding Site Profiles	HepG2		hg19_2
ATF3	ENCODE Transcription Factor Binding Site Profiles	K562		hg19_2
NR0B1	CHEA Transcription Factor Binding Site Profiles	MESC	Mouse	
CREB1	CHEA Transcription Factor Binding Site Profiles	GC1-SPG	Mouse	
FOXP1	CHEA Transcription Factor Binding Site Profiles	HESC	Human	
POU5F1	CHEA Transcription Factor Targets			
LMO2	CHEA Transcription Factor Targets			

SREBF2 has 21 predicted transcription factor associations.

Predicted Association	**Dataset**
RELA	JASPAR Predicted Transcription Factor Targets
NFYB	JASPAR Predicted Transcription Factor Targets
IRF2	JASPAR Predicted Transcription Factor Targets
TFAP2C	JASPAR Predicted Transcription Factor Targets
TEF-1	MotifMap Predicted Transcription Factor Targets
MAFA	MotifMap Predicted Transcription Factor Targets
ETS2	MotifMap Predicted Transcription Factor Targets
NURR1	MotifMap Predicted Transcription Factor Targets

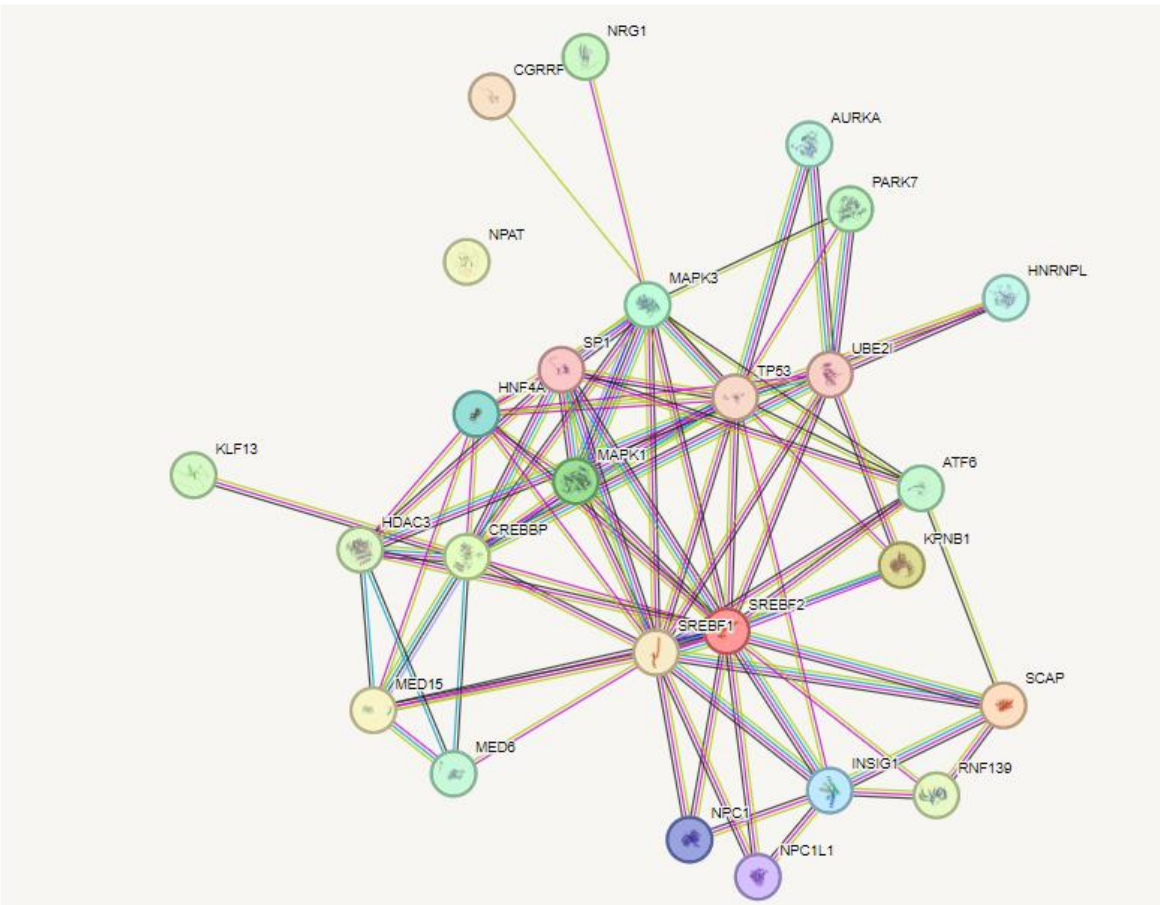
Predicted Association	Dataset
YY1	MotifMap Predicted Transcription Factor Targets
NF-Y	MotifMap Predicted Transcription Factor Targets
LBP-1	MotifMap Predicted Transcription Factor Targets
FXR	MotifMap Predicted Transcription Factor Targets
Neuro D	MotifMap Predicted Transcription Factor Targets
Nrf-1	MotifMap Predicted Transcription Factor Targets
SOX10	MotifMap Predicted Transcription Factor Targets
PKNOX2	MotifMap Predicted Transcription Factor Targets
45566	MotifMap Predicted Transcription Factor Targets
LRF	MotifMap Predicted Transcription Factor Targets
Octamer	MotifMap Predicted Transcription Factor Targets
STAT4	MotifMap Predicted Transcription Factor Targets
SOX4	MotifMap Predicted Transcription Factor Targets

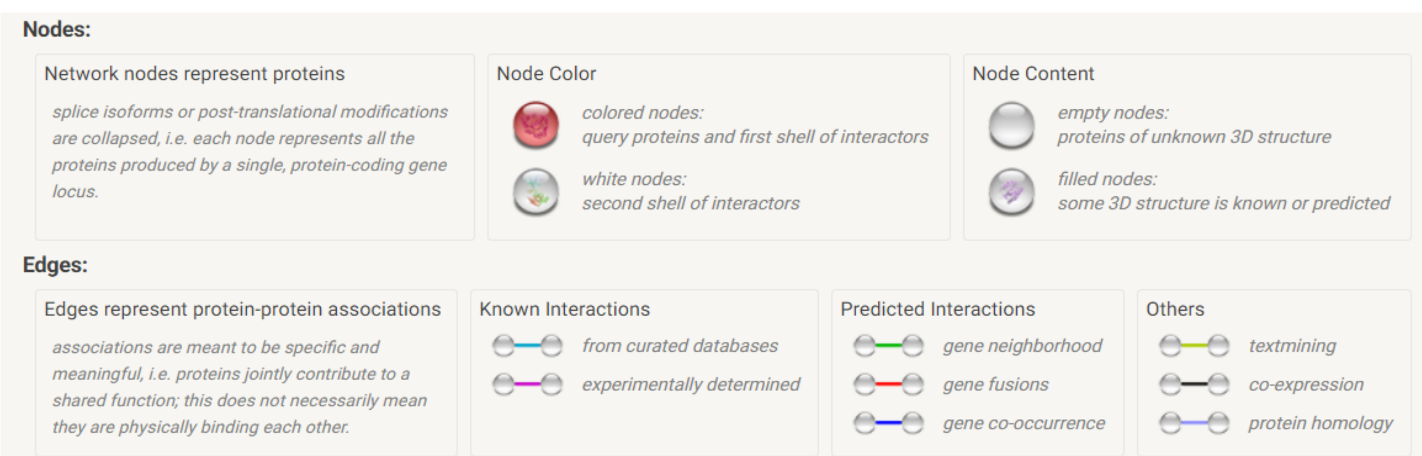
The download of all transcription factor associations is available as Supplementary Material (SREBF2_Harmonizome.xlsx).

5. Protein-protein interaction (PPI) partners

a. STRING

STRING is a database of known and predicted protein-protein interactions (PPIs). The interactions include direct (physical) and indirect (functional) associations; they stem from computational prediction, from knowledge transfer between organisms, and from interactions aggregated from other (primary) databases. Data sources include genomic context predictions, high-throughput lab experiments, (conserved) co-expression, automated text-mining, and previous knowledge in databases. The download of all interactions is available as Supplementary Material (SREBF2_STRING.xlsx)

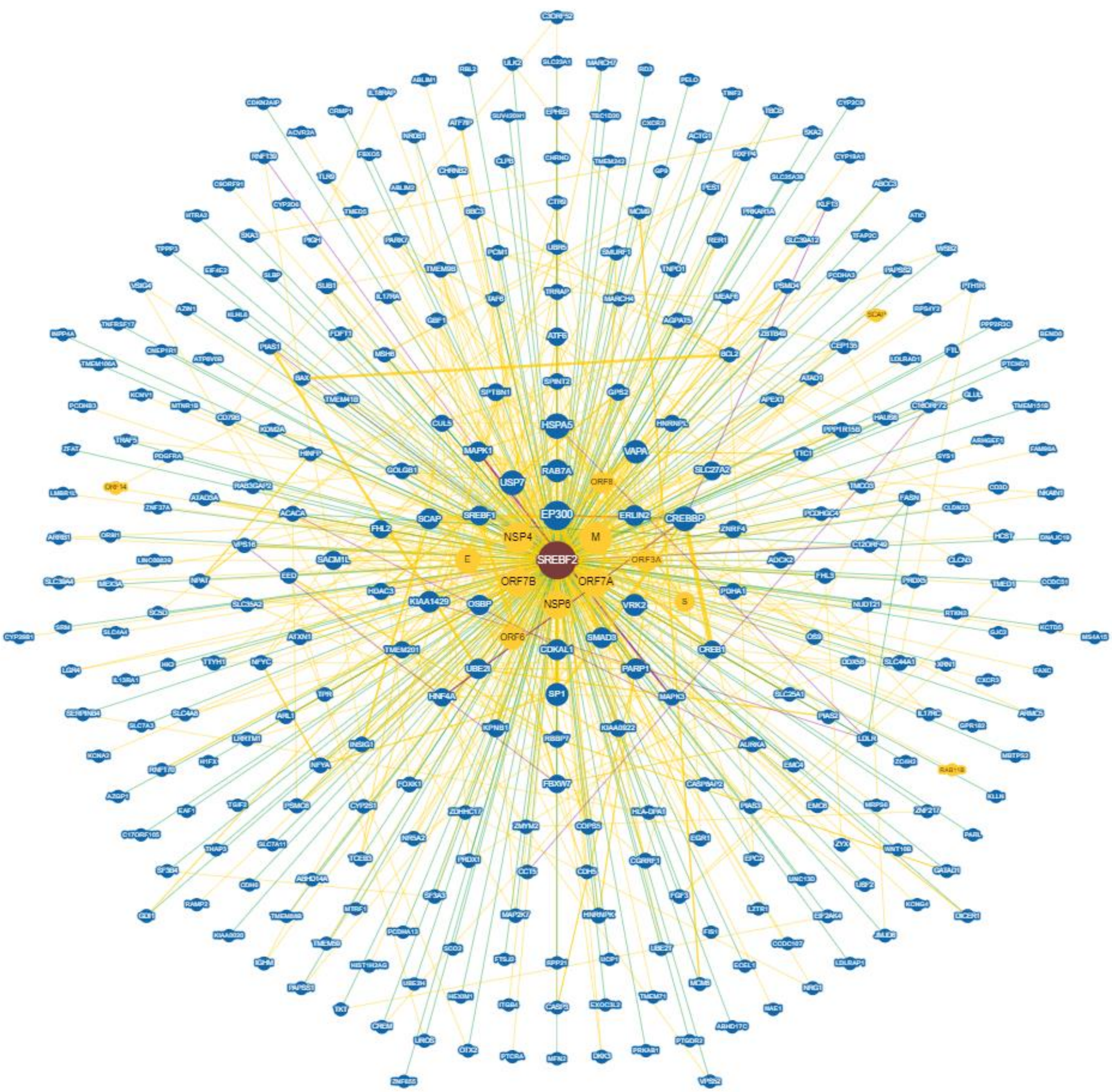




b. BioGRID

BioGRID is an interaction repository with data compiled through comprehensive curation efforts. The download of all interactions is available as Supplementary Material (SREBF2_BIOGRID.xlsx).

SREBF2 has 325 unique interactions.

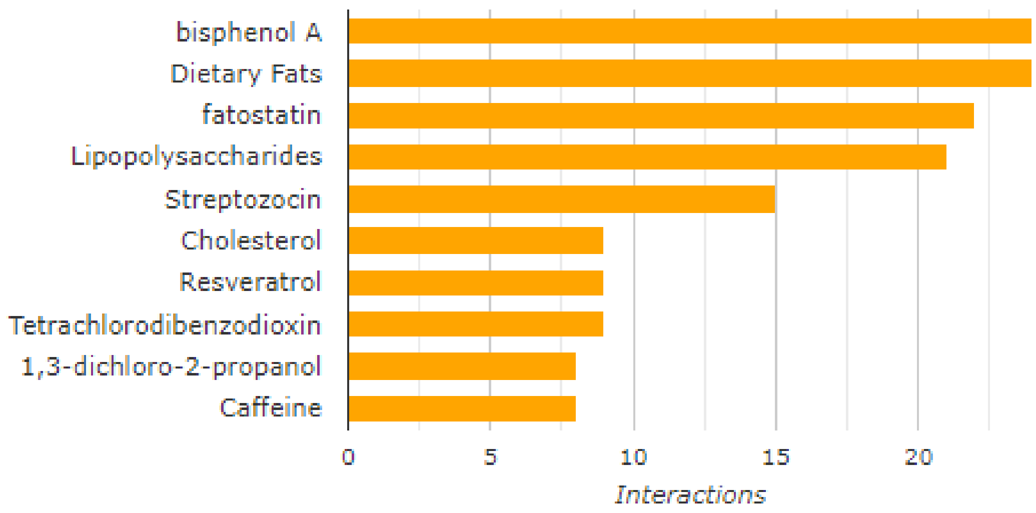


6. Information on which perturbations affect the gene

a. Comparative Toxicogenomics Database

Comparative Toxicogenomics Database (CTD) provides manually curated information about chemical–gene/protein interactions, chemical–disease and gene–disease relationships. The full list of interacting chemicals is provided as Supplementary Material (SREBF2_CTD.xlsx).

Top Interacting Chemicals



7. Available chemical tools and their effects on the gene

a. Queried in the following databases to identify small molecule inhibitors

SREBF2 gene was queried in the following databases to identify any known inhibitors or agonists and their effect(s) on the gene.

Chembl –

Please refer to the CDH10_Chembl_Inhibitor.xlsx file for the full records downloaded from Chembl.

DrugBank – No results

PubMed –

SREBP2 (the protein encoded by SREBF2 gene) inhibition by **fatostatin** impaired osteoclast differentiation in vitro and preserved bone mass and structure in a mouse RANKL-induced bone loss model. [Inoue and Imai, 2015]

Betulin is identified as a small molecule inhibitor of the SREBP-2 transcription factor. It efficiently reduces necrotic cell death induced by l-norephedrine in both SH-SY5Y neuroblastoma cells and primary cultured striatum neurons, by abrogating the activation of SREBP-2. [Funakoshi at al., 2016]

Simvastatin, an FDA-approved cholesterol-lowering drug, was shown to suppress hepatocellular carcinoma (HCC) tumor growth and sensitize HCC cells to sorafenib by inhibiting SREBP2-mediated cholesterol biosynthesis. [Mok et al., 2022]

Curcumin inhibits the proteolytic process of SREBP-2 in Caco-2 cells by downregulating the mRNA and protein expression of site-1 protease (S1P) rather than directly affecting SREBP-2 expression. This results in reduced production of mature SREBP-2 (mSREBP-2) but does not significantly impact the expression of precursor SREBP-2 (pSREBP-2) SR for at least a short period. [Li and Wu, 2020]

Synthesis and evaluation of **diarylthiazole derivatives** that inhibit activation of sterol regulatory element-binding proteins. [Kamisuki et al., 2011]

The diarylthiazole derivative, now called fatostatin, impairs the activation process of SREBPs, thereby decreasing the transcription of lipogenic genes in cells. **Fatostatin** (125B11), a specific inhibitor of SREBP activation, impairs the activation of SREBP-1 and SREBP-2. **Fatostatin hydrobromide** binds to SCAP (SREBP cleavage-activating protein) and inhibits the ER-Golgi translocation of SREBPs. [Kamisuki et al, 2009]

This publcation shows that **PCSK9-IN-9** and **5-O-methylembelin** could inhibit PCSK9, inducible degrader of the low-density lipoprotein receptor (IDOL), and SREBP2 mRNA expression. [Pel at al., 2022]

PubChem –

PubChem AID	BioAssay Name	Encoding Gene	Comment
615678	Inhibition of SREBP2 activation expressed in CHO-K1 cells assessed as decrease in mature form of SREBP at 1 to 20 uM after 6 hrs by Western blot analysis	SREBF2	Fatostatin, a recently discovered small molecule that inhibits activation of sterol regulatory element-binding protein (SREBP), blocks biosynthesis and accumulation of fat in obese mice. We synthesized and evaluated a series of fatostatin derivatives. Our structure-activity relationships led to the identification of N-(4-(2-(2-propylpyridin-4-yl)thiazol-4-yl)phenyl)methanesulfonamide (24, FGH10019) as the most potent druglike molecule among the analogues tested. Compound 24 has high aqueous solubility and membrane permeability and may serve as a seed molecule for further development.
1289613	Inhibition of SREBP2 maturation in human HepG2 cells at 25 uM after overnight incubation by immunoblot analysis in the presence of atorvastatin	SREBF3	Synthesis and Characterization of Fatty Acid Conjugates of Niacin and Salicylic Acid [PMID: 26784936]. The fatty acid niacin conjugate 5 has been shown to be an inhibitor of the sterol regulatory element binding protein (SREBP), a key regulator of cholesterol metabolism proteins such as PCSK9, HMG-CoA reductase, ATP citrate lyase, and NPC1L1.

NCATS Inxight: Drugs – No results

b. Tractability assessments for small molecule

Small molecule tractability – Druggable Family

8. eQTLs that alter the gene expression in the brain tissue

a. GTEx eQTL Browser

An expression quantitative trait locus (eQTL) is a locus that explains a fraction of the genetic variance of a gene expression phenotype, and can be thought of as a 'molecular link' between genetic variation and organismal phenotypes. A search of the GTEx portal identified 748 significant eQTLs for SREBF2 gene for all tissues with 42 of them where significant in brain tissues. The list of eQTLs in the brain is provided as Supplementary Material (SREBF2_GTEX_eQTL.xlsx).

9. Known SNPs linked to function/disease

a. ClinVar

ClinVar is a freely accessible resource of archived and aggregated information that can be utilized to evaluate relationships between human sequence variation and phenotypes. A search in ClinVar produced 123 results, 21 of which are pathogenic. ClinVar reports a link of SREBF2 gene with Inborn genetic diseases and Adenylosuccinate lyase deficiency. The full list of results is available as Supplementary Material (SREBF2_ClinVar.xlsx).

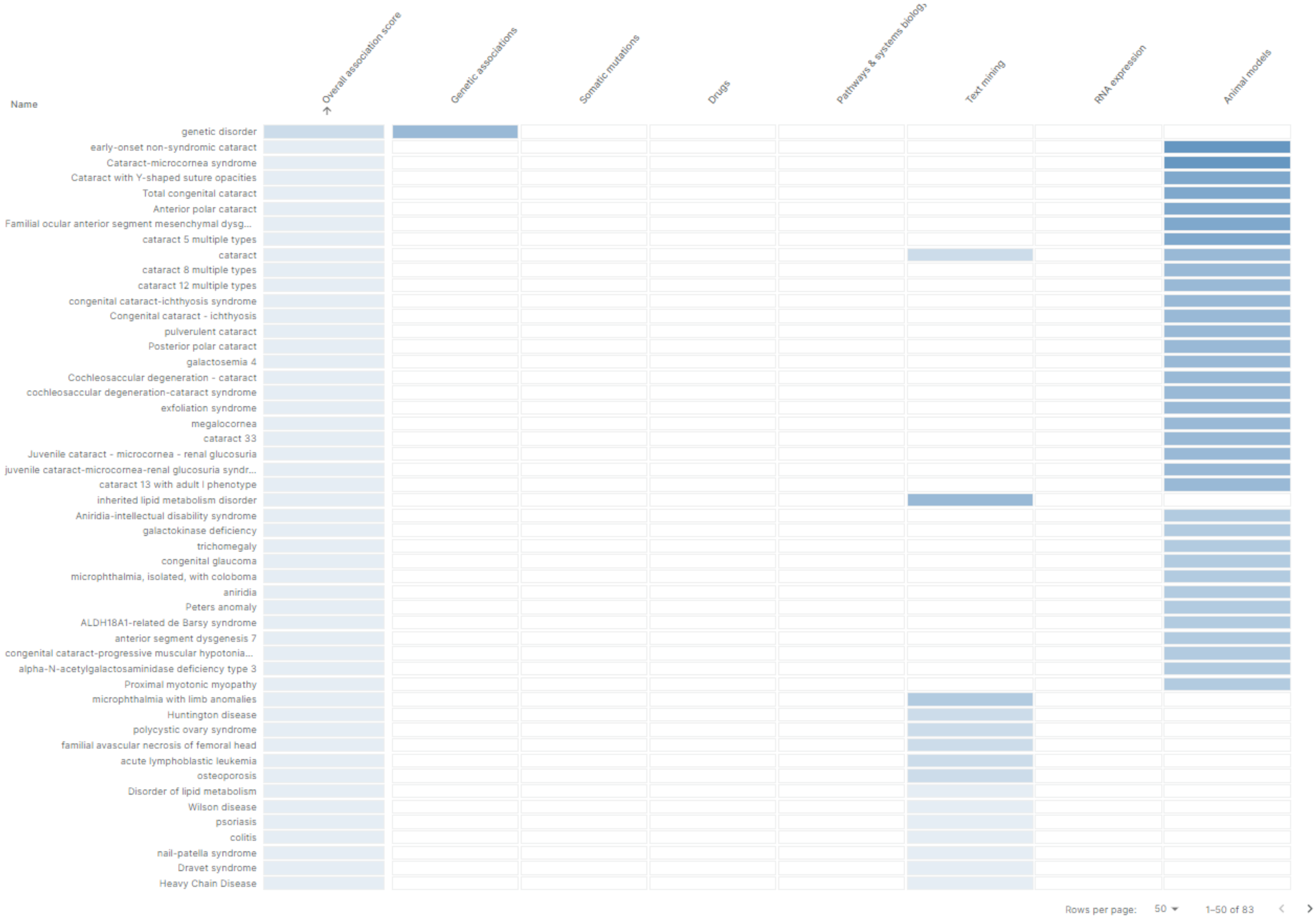
b. OMIM

OMIM (Online Mendelian Inheritance in Man) is a freely available resource that compiles information focused on the relationship between phenotype and genotype. OMIM describes a link of MSH3 gene with hypercholesterolemia in man.

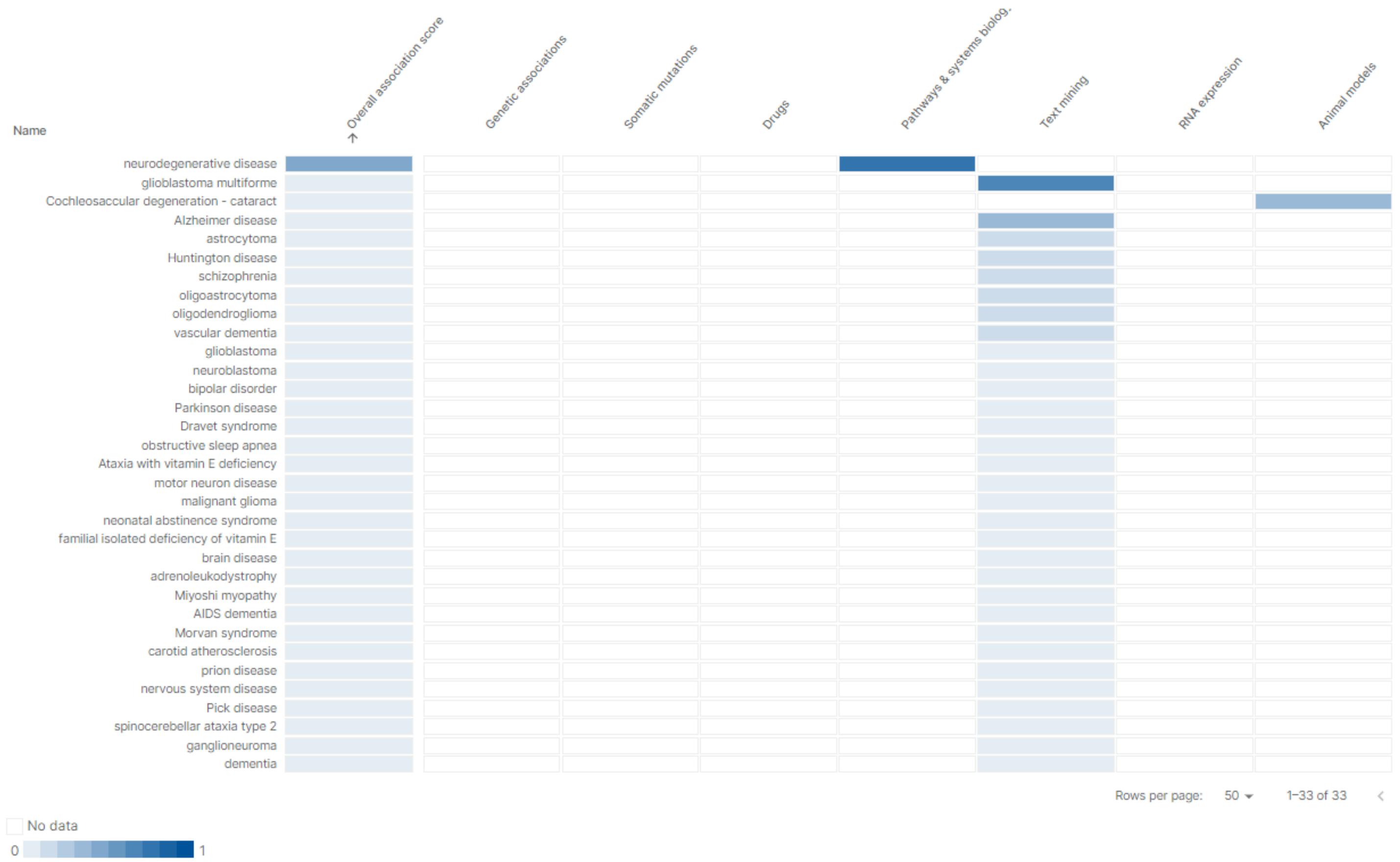
c. Open Targets

The Open Targets Platform is a resource that enables query and visualization of potential drug targets associated with disease, with the goal of helping to prioritize targets for further investigation. A search of SREBF2 in Open Target revealed 286 disease associations including genetic (n=83, top 50 are shown below) and nervous system disease (n=33) associations.

Genetically linked with disease (top 50 was shown)



Nervous system diseases



b. SNPs3D

SNPs3D is a website which assigns molecular functional effects of non-synonymous SNPs based on structure and sequence analysis. The corresponding data were generated by the John Moult's Group (Link). A query of SNPs3D database found 8 SNPs related to the gene of interest (SNP analysis on Disease Gene: SREBF2 (sterol regulatory element binding transcription factor 2)).

dbSNP

refseq accession	snp	snp_id	svm profile	svm structure	molecular effect	model frequency role
NP_004590						
	Q308H	11547823	-0.12			
	K310E	11547822	-2.40			
	M536L	17002714	2.10			0.00
	A592T	6002520	-0.33			
	G595A	4822063	1.23			0.39
	V623M	2229440	0.71			0.03
	R860S	11547820	1.97			
	V902L	17848351	-1.18			

10. Major pathways

a. Pathways associated with the gene

Reactome pathways

- Activation of gene expression by SREBF (SREBP)
- Cholesterol biosynthesis
- EGR2 and SOX10-mediated initiation of Schwann cell myelination
- Nervous system development
- PPARA activates gene expression
- Transcriptional regulation of white adipocyte differentiation

WikiPathways

- 22q11.2 copy number variation syndrome
- Angiopoietin-like protein 8 regulatory pathway
- Cholesterol metabolism with Bloch and Kandutsch-Russell pathways
- SREBF and miR33 in cholesterol and lipid homeostasis
- Sterol regulatory element-binding proteins (SREBP) signaling

11. Available knockouts of the gene and their phenotype

a. PubMed

Sterol regulatory element binding proteins (SREBPs) are a family of transcription factors that regulate lipid homeostasis [Eberle et al., 2004]. The three SREBP isoforms in mammals are designated SREBP-1a, SREBP-1c, and SREBP-2 (encoded by SREBF2 gene). They belong to the basic helix-loop-helix leucine zipper (bHLH-Zip) family of transcription factors [Goldstein et al., 2002]. In vivo studies using transgenic and knockout mice suggest that SREBP-1c is involved in fatty acid synthesis and insulin induced glucose metabolism (particularly in lipogenesis), whereas SREBP-2 is relatively specific to cholesterol synthesis [Eberle et al., 2004]. SREBP-2 can replace SREBP-1 in regulating cholesterol synthesis in livers of mice and that the higher potency of SREBP-2 relative to SREBP-1c leads to excessive hepatic cholesterol synthesis [Shimano et al., 1997]. Mice deficient in SREBP-2 (Srebf2-/-) showed embryonic lethality, limb development disruption, and dysregulation of morphogenic gene expression. A single surviving Srebf2-/- mouse exhibited alopecia, stunted growth, and diminished adipose tissue. In contrast, hypomorphic mice with low SREBP-2 levels experienced gender-specific outcomes: females had reduced body weight and succumbed between 8 to 12 weeks, while males survived with decreased hepatic cholesterol and downregulated SREBP target genes [Vergnes et al., 2016].

Pharmacological inhibition of SREBP2 or Srebf2 RNA interference attenuated mouse liver cancer initiation induced by gut flora disequilibrium [Chen et al., 2022]. Mice with knockout of SREBP2 in astrocytes have impaired brain development and behavioral and motor defects. These mice also have altered energy balance, altered body composition, and a shift in metabolism toward carbohydrate oxidation driven by increased glucose oxidation by the brain. Thus, SREBP2-mediated cholesterol synthesis in astrocytes plays an important role in brain and neuronal development and function, and altered brain cholesterol synthesis may contribute to the interaction between metabolic diseases, such as diabetes and altered brain function [Ferris et al., 2017].

b. Mouse Genome Informatics (MGI)

The Mouse Genome Informatics (MGI) database is a comprehensive resource that integrates genetic, genomic, and biological data pertaining to the laboratory mouse. A search in MGI identified 30 gene trapped, 2 targeted and 2 radiation induced mutations for SREBF2 (SREBF2_MGI.xlsx).

Strain ID	Strain/Stock	Repository	State	Allele Name	URL
JAX:001385	STOCK Srebf2<lop13>/J	JAX	embryo	lens opacity 13	link
JAX:001385	STOCK Srebf2<lop13>/J	JAX	embryo	lens opacity 13	link
JAX:030826	STOCK Srebf2<tm1Jdh>/J	JAX	sperm	targeted mutation 1, Jay D Horton	link

JAX:031792	STOCK Srebf2<tm1.1Jdh>/J	JAX	sperm	targeted mutation 1.1, Jay D Horton	link
NM-KO-210188	C57BL/6Smoc-Srebf2<emSmoc>	SMOC	embryo		link
NM-CKO-200136	C57BL/6Smoc-Srebf2<em1(flox)Smoc>	SMOC	embryo		link
CARD:1525	B6;CB-Srebf2<Gt(pU-21W)399Card>	CARD	embryo		link

Additionally, 15 phenotypes from 3 alleles in 4 genetic backgrounds have been reported for SREBF2. The full list of results is available as Supplementary Material (SREBF2_MGI.xlsx).

Allelic Composition	Genetic Background	Annotated Term	Phenotype Summary Category	Reference
Srebf2<Gt(YTA054)Byg>/Srebf2<Gt(YTA054)Byg>	involves: 129P2/OlaHsd * C57BL/6J	prenatal lethality, complete penetrance	mortality/aging	J:176249
Srebf2<Gt(YTA054)Byg>/Srebf2<lop13>	involves: 129P2/OlaHsd * 129T1/Sv * C57BL/6J	prenatal lethality, incomplete penetrance	mortality/aging	J:176249
Srebf2<lop13>/Srebf2<lop13>	STOCK Srebf2<lop13>/J	decreased brain cholesterol level	nervous system	J:176249

12. Major labs working on the gene based on publications

Dr. Goldstein JL
Department of Molecular Genetics, University of Texas Southwestern Medical Center, Dallas, TX 75235, USA.
Web site: https://profiles.utsouthwestern.edu/profile/12645/joseph-goldstein.html?_ga=2.212222676.1346133475.1710549430-840369987.1710549430&
Pubmed link

Dr. Helen H. Hobbs and Dr. Jonathan C. Cohen
Department of Internal Medicine, University of Texas Southwestern Medical Center, Dallas 75235, USA.
Email: jonathan.cohen@utsouthwestern.edu
Web site: https://labs.utsouthwestern.edu/hobbs-cohen-lab
Pubmed link

Dr. Anna Colell Riera
Department of Cell Death and Proliferation, Institut d'Investigacions Biomèdiques de Barcelona (IIBB-CSIC)
Email: anna.colell@iibb.csic.es
Web site: https://www.iibb.csic.es/en/research/61
Pubmed

Dr. Hitoshi Shimano
Department of Endocrinology and Metabolism, Faculty of Medicine, University of Tsukuba, Tsukuba, Japan
Email: hshimano@md.tsukuba.ac.jp
Web site: https://www.md.tsukuba.ac.jp/gradmed/data/research/Metabolism_and_Endocrinology.pdf
Pubmed

13. Commercial antibodies

Antibody product name	Vendor name	Catalog number	Antibody description
Anti-SREBP2 antibody (ab30682)	Abcam	ab30682	Rabbit polyclonal to SREBP2
Anti-SREBP2 antibody (ab28482)	Abcam	ab28482	Rabbit polyclonal to SREBP2
Anti-SREBP2 antibody (ab112046)	Abcam	ab112046	Rabbit polyclonal to SREBP2
Anti-SREBP2 antibody (ab194667)	Abcam	ab194667	Mouse polyclonal to SREBP2
SREBP-2 Antibody (1C6)	Santa Cruz	sc-13552	mouse monoclonal IgG1 κ SREBP-2 antibody
SREBF2 antibody	biorbyt	orb592752	Rabbit polyclonal to SREBF2
SREBF2 antibody	SinoBiological	102143-T32	Rabbit polyclonal to SREBP2
SREBP2 Antibody (1D2) - BSA Free	Novus	NBP1-5446	Mouse polyclonal to SREBP2

SREBP2 Antibody	Novus	NB100-74543	Rabbit Polyclonal Unconjugated
SREBP2 Antibody	Novus	AF7119	Goat Polyclonal Unconjugated
SREBP2 Antibody (751512) [Unconjugated]	Novus	MAB7119	Mouse Monoclonal Unconjugated
SREBP2 Antibody	Novus	NBP1-71880	Rabbit Polyclonal Unconjugated
SREBP2 Antibody - BSA Free	Novus	NBP2-41282	Rabbit Polyclonal Unconjugated
SREBP2 Antibody (SREBP2/1580)	Novus	NBP3-13783	Mouse Monoclonal Unconjugated
SREBP2 Antibody (SREBP2/1579)	Novus	NBP3-07307	Mouse Monoclonal Unconjugated
SREBP2 Antibody	Novus	NBP2-58133	Rabbit Polyclonal Unconjugated
SREBP2 Antibody [mFluor Violet 500 SE]	Novus	NBP2-41282MFV500	Rabbit Polyclonal Conjugate:mFluor Violet 500 SE
SREBP2 Antibody (SREBP2/1579) [Alexa Fluor® 350]	Novus	NBP3-08527AF350	Mouse Monoclonal Conjugate:Alexa Fluor 350
Human SREBP2 Antibody	R&D systems	MAB7119	Monoclonal Mouse IgG1
Human SREBP2 Antibody	R&D systems	AF7119	Polyclonal Goat IgG
IHC-plus™ Polyclonal Rabbit anti-Human SREBF2 / SREBP2 Antibody (aa455-469, IHC, WB) LS-B1609	LSBio	LS-B1609	SREBF2 / SREBP2 Rabbit anti-Human Polyclonal (aa455-469) Antibody
Polyclonal Rabbit anti-Human SREBF2 / SREBP2 Antibody (aa304-353, IHC, WB) LS-B4695	LSBio	LS-B4695	SREBF2 / SREBP2 Rabbit anti-Human Polyclonal (aa304-353) Antibody

14. Information on Patents

Title	Patents	Applicant	Abstract/Summary
Method for the diagnosis/prognosis of colorectal cancer.	CA-2799359-C	Consejo Superior De Investigaciones Cientificas, Jose Ignacio Casal Alvarez, Rodrigo Barderas Manchado, Ingrid Henriette Suzanne Babel	The present invention relates antibody capturing entities (ACEs), and compositions, methods and kits containing or using same for the diagnosis, prognosis or monitoring of colorectal cancer (CRC) progression in a subject. The ACE is selected from amino acid sequences shown in SEQ ID NOS. 1-6, and variants and combinations thereof, wherein the ACEs are recognizable by anti-SULF1-, anti-GRN-, anti-GTF2i-, anti-MST1-, anti-SREBF2- and anti-NHSL1-autoantibodies. The methods involve contacting a sample from a subject with at least one ACE and detecting the formation of an autoantibody-ACE complex, wherein the detection of the autoantibody-ACE complex is indicative of CRC and/or progression thereof.
Nanogold-labeled chip for polygene diagnosis of prostate cancer	CN-102888450-A	陈清标, 陈锡彬, 钟惟德	The invention discloses a nanogold-labeled chip for polygene diagnosis of prostate cancer. The nanogold-labeled chip mainly comprises 11 genes, a prostate cancer cell strain RNA (Ribonucleic Acid), a nanogold-labeled oligonucleotide probe and nitrocellulose, wherein the 11 genes are obtained by selection of the gene chip and verification of fluorescent quantitation polymerase chain reaction (PCR), have obvious expression difference in prostate cancer and benign prostatic hyperplasia and include IGF1, P27, AR, PSMA, KLK3, PCNA, Ki-67, KLK1, KLK2, TMPRSS2 and SREBF2. The nanogold-labeled chip for polygene diagnosis of prostate cancer can be applied in clinically early diagnosis on prostate cancer.
Composition for preventing or treating infectious disease comprising SREBP2 inhibitors	KR-20220023276-A	서영교, 배종섭, 이원화, 최은영, 유영범, 안준홍	The present invention relates to a composition for preventing or treating infectious diseases comprising a SREBP2 inhibitor, and more particularly, to a composition for preventing or treating infectious diseases comprising a substance inhibiting the expression or activity of mRNA or protein of SREBP2 as an active ingredient, and to a method for screening a therapeutic agent for infectious diseases. The composition for preventing or treating infectious diseases comprising an inhibitor of SREBP2 expression or activity provided by the present invention suppresses the release of inflammatory cytokines and vascular barrier collapse according to infection, and exhibits an effect of improving a survival

			rate, thereby being able to be usefully used to develop therapeutic agents for infectious diseases.
Composition for diagnosing infectious inflammatory disease comprising agents detecting the expression level of SREBP2	KR-102573556-B1	서영교, 배종섭, 이원화, 최은영, 유영범, 안준홍	The present invention relates to a composition for diagnosing an infectious disease comprising an agent for measuring the expression level of SREBP2 (Sterol regulatory element binding proteins), and more particularly, the expression level of SREBP2 (Sterol regulatory element binding proteins) or its C-terminal peptide It relates to a composition for diagnosing an infectious disease or diagnosing its severity, including an agent for measuring. According to the present invention, since the expression level of SREBP2 or its C-terminal peptide increases in proportion to the severity of an infectious disease, these markers can be very useful for diagnosing and predicting the severity of an infectious disease.
Methods and compositions for treatment of aberrant hematopoiesis	US-2022008504-A1	Longhou FANGQilin Gu	Embodiments of the present disclosure pertain to methods of reducing hematopoiesis in a subject by administering to the subject a therapeutically effective amount of an inhibitor of Srebp2. Additional embodiments of the present disclosure pertain to compositions for reducing hematopoiesis in a subject. In some embodiments, the compositions include a therapeutically effective amount of an inhibitor of Srebp2. Further embodiments of the present disclosure pertain to methods of enhancing hematopoiesis in a subject by administering to the subject a therapeutically effective amount of an active ingredient. Additional embodiments of the present disclosure pertain to compositions for enhancing hematopoiesis in a subject. In some embodiments, the compositions of the present disclosure include active ingredients that enhance hematopoiesis in the subject.

15. NIH and ERC grants

a. NIH RePORTER

The NIH Research Portfolio Online Reporting Tools (RePORT) website provides access to reports, data, and analyses of NIH research activities, including information on NIH expenditures and the results of NIH-supported research. The RePORT Expenditures and Results (RePORTER) module identifies 3 active SREBF2-related grants funded by the NIH, which are shown below. A list of all SREBF2-related grants is available as Supplementary Material (SREBF2_NIH_RePORTER.xlsx).

Project Title	Project Number	Organization Name	Budget Start Date	Budget End Date	Funding IC(s)
The importance of Treg-intrinsic cholesterol metabolism for visceral adipose tissue Treg homeostasis, phenotype, and function	1F31DK137590-01	EMORY UNIVERSITY	7/1/2023	6/30/2024	NIDDK
Regulation of successful optic nerve regeneration by the mevalonate/cholesterol pathway	5R01EY034097-02	MEDICAL COLLEGE OF WISCONSIN	4/1/2023	3/31/2024	NEI
Site-1 protease-mediated lipid metabolism in lymphatic vascular development	5R01HL153728-04	OKLAHOMA MEDICAL RESEARCH FOUNDATION	6/1/2023	5/31/2024	NHLBI

b. ERC

The European Research Council (ERC) is the premier European funding organization for excellent frontier research. It funds creative researchers of any nationality and age, to run projects based across Europe. It encourages the highest quality research in Europe through competitive funding and to support investigator-driven frontier research across all fields, based on scientific excellence.
No SREBF2-related grant funded by the ERC was found.

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Compiled List of Relevant Databases

1. GeneCards: https://www.genecards.org/
2. USCS Genome Browser: https://genome.ucsc.edu/
3. NCBI Structure Group: https://www.ncbi.nlm.nih.gov/Structure/index.shtml
4. The RCSB Protein Data Bank: https://www.rcsb.org/
5. AlphaFold Protein Structure Database: https://www.alphafold.ebi.ac.uk/
6. NCBI Annotation Pipeline: https://www.ncbi.nlm.nih.gov/genome/annotation_euk/process/
7. GTEx Portal: https://gtexportal.org/home/
8. The Human Brain Transcriptome (HBT) Project: http://hbatlas.org/
9. Allen Brain Map: http://portal.brain-map.org/
10. Barres Lab Brain RNA-Seq Portal: http://www.brainrnaseq.org/
11. DropViz: http://dropviz.org/
12. NCBI GEO Profiles: https://www.ncbi.nlm.nih.gov/geoprofiles/
13. Harmonizome: https://maayanlab.cloud/Harmonizome/
14. STRING: https://string-db.org/
15. BioGRID: https://thebiogrid.org/
16. Comparative Toxicogenomics Database: http://ctdbase.org/
17. Chembl: https://www.ebi.ac.uk/chembl/
18. DrugBank: https://www.drugbank.ca/
19. PubMed: https://www.ncbi.nlm.nih.gov/pubmed/

20. PubChem: https://pubchem.ncbi.nlm.nih.gov/
21. NCATS Inxight: https://drugs.ncats.io/about
22. NCBI ClinVar: https://www.ncbi.nlm.nih.gov/clinvar/
23. Online Mendelian Inheritance in Man (OMIM): https://www.omim.org/
24. Open Targets: https://www.targetvalidation.org/
25. Pathway Commons: https://apps.pathwaycommons.org/
26. Reactome: https://reactome.org/
27. Mouse Genomic Informatics (MGI): http://www.informatics.jax.org/
28. NIH Research Portfolio Online Reporting Tools (RePORT): https://report.nih.gov/
29. European Research Council (ERC): https://erc.europa.eu/homepage